

# AMERICAN CARRIER STRIKE GROUPS: An Electronic Perspective



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In the last four parts of this article we covered air defence warfare, electronic warfare (EW), under-sea warfare, strike warfare and air operations. In this concluding part, let us take a look at command and control (C<sup>2</sup>) operations as well as cruise control operations.

## Radars—A revisit

All surveillance and navigation radars are situated high up in the mast. In the age of sail ships, observers used to sit high up in the mast. Who can forget the observers ringing the bell after seeing the iceberg in the movie *Titanic*? Radars sitting high up in the masts give a greater radar horizon. Radar horizon is the maximum range at which radar can detect a target against the curvature of earth.

But presence of so many communication and detection equipment in the mast in interference. This requires

certain systems to be turned off when another system is working. If the turned-off system is an important system, and if the enemy knowingly or by chance chooses to attack during this window, the result can be catastrophic.

## C<sup>2</sup> operations

Whatever may be the strength of force, it is only the right commands given at right times can make the force lethal. The commander who commands the carrier strike group (CSG) must know what is happening around him, the strengths and the weaknesses of the group.

For example, what will happen if the commander issues a command to launch a surface-to-air missile (SAM) when stocks of SAM have been exhausted? He must know the threats and his capabilities, and must be present in the carrier to command the CSG.

For effective command, the commander requires an efficient C<sup>2</sup> system. Backbone of C<sup>2</sup> systems is the communication systems that include voice links and datalinks.

## Communication systems

The mast of the carrier is littered with numerous antennae—protected by radomes—for various communication systems and datalinks present in the CSG. Almost all sorts of antenna systems are seen in the carrier right from whip antennae situated in the flight deck to wire-fed fan antennae present above the superstructure. A plethora of communication systems operating at various frequencies and following various techniques of transmission could be found in the CSG.

**Terrestrial radio communication.** It has both military and non-military radio communication facilities operating in MF, HF, VHF and UHF bands. Military communications are secured through encryption



**Fig. 30:** Radars in the mast of the carrier. The reader may try to identify the radars present in the masts of the carrier. This image covers many radars and also an ECM module (Image courtesy: US Navy)